

Gauss's Law, solved

Step-by-step solutions to every practice problem in Chapter 14. Try each problem on your own before you read its solution.

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P14.1

No, the field need not be zero. Zero net flux only means that as much flux leaves the surface as enters it. For example, a closed surface in a uniform external field has field passing through it everywhere, yet zero net flux (Example 14.1).

No, the surface need not be empty. Gauss's law gives zero net flux whenever the *net* enclosed charge is zero. The surface could contain $+q$ and $-q$, such as a whole dipole.

P14.2

Gauss's law holds for any closed surface, but it gives only the *integral* $\oint \vec{E} \cdot d\vec{A}$. To extract $\vec{E}$ from that integral, we need a surface on which $\vec{E}$ is constant in magnitude and either perpendicular to the surface or parallel to it. Then the integral is just $\vec{E}$ times an area.

A uniformly charged sphere has spherical symmetry, so its field is radial and has the same size everywhere on a concentric sphere. A charged cube has no such symmetry. Its field varies in both magnitude and direction over any simple surface, so we cannot pull $\vec{E}$ out of the integral. Its field must be found by integrating Coulomb's law, or numerically.

P14.3

The car's metal body acts as a (leaky) **Faraday cage**. The lightning current flows over the outer metal skin, and charges rearrange on the conductor so that the field inside the passenger compartment stays very small. The current then jumps from the body to the ground, often

through the wet road or across the tyres. The protection is **not** due to the rubber tyres: a few centimetres of rubber cannot stop a discharge that has just crossed kilometres of air.

A convertible with a fabric roof lacks a continuous metal enclosure overhead, so it protects its occupants much less well. Golf carts and motorbikes offer essentially no protection.

P14.4

It moves to the outer surface. Inside the metal, $\mathbf{E} = \mathbf{0}$ in equilibrium. A Gaussian surface drawn just inside the surface then encloses no net charge, so all the excess charge must lie on the surface.

An egg shape. The charge is *not* spread uniformly. The surface charge density, and so the field just outside (σ/ϵ_0), is greatest where the surface is most sharply curved, at the pointed end. This is why charge leaks away from sharp points, and why lightning rods are pointed.

P14.5

Close the hemisphere with its flat circular base. That closed surface encloses no charge, so the total flux through it is zero.

The base is perpendicular to the field, so the field crosses it with flux of magnitude $E\pi R^2$. The same flux must cross the curved surface in the opposite sense. Taking the outward normal, with the field entering through the base and leaving through the curved surface:

$$\Phi_{\text{curved}} = +E\pi R^2 = (2.0 \times 10^4)\pi(0.10)^2 = 6.3 \times 10^2 \text{ N m}^2/\text{C}.$$

The flux through the curved surface equals the flux through its “shadow”, the disc it covers.

P14.6

Charge at the centre. The total flux is q/ϵ_0 , and by symmetry it divides equally among the six faces:

$$\Phi_{\text{face}} = \frac{q}{6\epsilon_0} = \frac{3.0 \times 10^{-9}}{6(8.854 \times 10^{-12})} = 56 \text{ N m}^2/\text{C}.$$

Charge at a corner. Imagine eight identical cubes meeting at that corner, together forming one large cube with the charge at its centre. Each small cube receives one-eighth of the total flux,

$$q/(8\epsilon_0) = 42 \text{ N m}^2/\text{C}.$$

Within our cube:

- The three faces that meet at the charge receive **zero** flux, because the field runs along them.
- The flux therefore divides among the other three faces, $q/(24\epsilon_0) = 14 \text{ N m}^2/\text{C}$ each.

P14.7

- At 0.20 m (inside the shell): $Q_{\text{enc}} = 0$, so $E = 0$.
- At 0.50 m (outside):

$$E = \frac{kQ}{r^2} = \frac{(8.99 \times 10^9)(6.0 \times 10^{-9})}{0.25} = 216 \text{ N/C, radially outward.}$$

P14.8

Inside ($r < R$). Use a coaxial Gaussian cylinder of radius r and length ℓ :

$$E(2\pi r\ell) = \frac{\rho\pi r^2\ell}{\epsilon_0} \Rightarrow E = \frac{\rho r}{2\epsilon_0} = \frac{(5.0 \times 10^{-6})(0.010)}{2(8.854 \times 10^{-12})} = 2.8 \times 10^3 \text{ N/C}.$$

Outside ($r > R$). The enclosed charge is $\rho\pi R^2\ell$:

$$E = \frac{\rho R^2}{2\epsilon_0 r} = \frac{(5.0 \times 10^{-6})(0.020)^2}{2(8.854 \times 10^{-12})(0.040)} = 2.8 \times 10^3 \text{ N/C}.$$

Comment. The two fields are equal. Inside, $E \propto r$, and outside, $E \propto 1/r$, so the field at r_1 (inside) equals the field at r_2 (outside) whenever $r_1 r_2 = R^2$. Here $1.0 \times 4.0 = 2.0^2$ (in cm^2). The field is greatest at the surface, $r = R$.

P14.9

Put sheet A ($+3.0 \mu\text{C}/\text{m}^2$) on the left and sheet B ($-1.0 \mu\text{C}/\text{m}^2$) on the right, and take $+x$ to the right. Each sheet produces a uniform field of magnitude $\sigma/2\epsilon_0$:

- Sheet A: $1.69 \times 10^5 \text{ N/C}$, pointing away from A.
- Sheet B: $0.565 \times 10^5 \text{ N/C}$, pointing towards B.

REGION	FIELD FROM A	FIELD FROM B	NET FIELD
Left of A	-1.69×10^5	$+0.565 \times 10^5$	1.1×10^5 N/C, to the left
Between	$+1.69 \times 10^5$	$+0.565 \times 10^5$	2.3×10^5 N/C, to the right
Right of B	$+1.69 \times 10^5$	-0.565×10^5	1.1×10^5 N/C, to the right

Outside the pair, the field corresponds to the net charge $+2.0 \mu\text{C}/\text{m}^2$: $\sigma_{\text{net}}/2\epsilon_0 = 1.13 \times 10^5$ N/C, pointing away from the pair. ✓

P14.10

Surface charges. The field is zero inside the metal, so the inner surface carries -3.0 nC , cancelling the central charge. The shell is neutral overall, so the outer surface carries $+3.0 \text{ nC}$.

Field $E(r)$:

- $r < 4 \text{ cm}$: $E = kq/r^2$, outward. For example, 6.7×10^4 N/C at 2 cm, and 1.7×10^4 N/C just inside 4 cm.
- $4 < r < 6 \text{ cm}$: $E = 0$, inside the metal.
- $r > 6 \text{ cm}$: $E = kq/r^2$ again, outward: 7.5×10^3 N/C at 6 cm, falling to 2.7×10^3 N/C at 10 cm.

Sketch. The curve is a $1/r^2$ decline from the centre, a gap where $E = 0$ inside the metal, then the same $1/r^2$ curve resuming at the outer surface. The shell hides *where* the charge is, but not *how much* there is.

P14.11

The field just outside a conductor is $E = \sigma/\epsilon_0$, and it points away from a positive surface. The field here points *down*, towards the surface, so the Earth's surface charge is negative:

$$\sigma = -\epsilon_0 E = -(8.854 \times 10^{-12})(100) = -8.9 \times 10^{-10} \text{ C/m}^2.$$

Total charge:

$$Q = \sigma \cdot 4\pi R_E^2 = (-8.9 \times 10^{-10})(4\pi)(6.37 \times 10^6)^2 = -4.5 \times 10^5 \text{ C}.$$

Context. This charge leaks away through the slightly conducting atmosphere in minutes. It is continually recharged by the roughly 2000 thunderstorms active around the world at any moment. This system is the global atmospheric electric circuit.

P14.12

For a sphere, the surface field is $E = kQ/R^2$:

$$Q = \frac{ER^2}{k} = \frac{(2.0 \times 10^6)(0.25)^2}{8.99 \times 10^9} = 1.4 \times 10^{-5} \text{ C}.$$

$$\sigma = \epsilon_0 E = (8.854 \times 10^{-12})(2.0 \times 10^6) = 1.8 \times 10^{-5} \text{ C/m}^2.$$

P14.13

(a) **Total charge.** Add up shells:

$$Q = \int_0^R \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr = 4\pi \rho_0 \left(\frac{R^3}{3} - \frac{R^3}{4}\right) = \frac{\pi \rho_0 R^3}{3}.$$

(b) **Field.** The charge inside radius r is

$$Q_{\text{enc}} = 4\pi \rho_0 \left(\frac{r^3}{3} - \frac{r^4}{4R}\right).$$

Gauss's law then gives:

- Inside: $E = \frac{Q_{\text{enc}}}{4\pi \epsilon_0 r^2} = \frac{\rho_0}{\epsilon_0} \left(\frac{r}{3} - \frac{r^2}{4R}\right).$
- Outside: $E = \frac{kQ}{r^2} = \frac{\rho_0 R^3}{12\epsilon_0 r^2}.$

Check at $r = R$: the inside formula gives $\frac{\rho_0}{\epsilon_0} \left(\frac{R}{3} - \frac{R}{4}\right) = \frac{\rho_0 R}{12\epsilon_0}$, which matches the outside formula. ✓

(c) **Maximum field.** Set $dE/dr = 0$:

$$\frac{1}{3} - \frac{r}{2R} = 0 \Rightarrow r = \frac{2R}{3}.$$

$$E_{\max} = \frac{\rho_0}{\epsilon_0} \left(\frac{2R}{9} - \frac{R}{9} \right) = \frac{\rho_0 R}{9\epsilon_0}.$$

The maximum lies *inside* the sphere, because the charge density falls to zero at the surface.

P14.14

Field inside a uniform sphere. At position $\vec{r}$ measured from the centre of a uniformly charged sphere of density ρ , Gauss's law gives $\vec{E} = \rho\vec{r}/3\epsilon_0$, pointing radially:

$$\vec{E} = \frac{\rho}{3\epsilon_0} \vec{r}.$$

Superposition. The sphere with a cavity is the same as a full sphere of density $+\rho$ plus a small sphere of density $-\rho$ filling the cavity. Take a point inside the cavity at position $\vec{r}$ from the big sphere's centre, which is $\vec{r} - \vec{a}$ from the cavity's centre:

$$\vec{E} = \frac{\rho}{3\epsilon_0} \vec{r} + \frac{(-\rho)}{3\epsilon_0} (\vec{r} - \vec{a}) = \frac{\rho}{3\epsilon_0} \vec{a}.$$

The result does not depend on $\vec{r}$, so the field is **uniform** throughout the cavity, parallel to the line joining the two centres. This is a classic example of the power of superposition.

P14.15

By symmetry, the field is perpendicular to the slab, zero on the mid-plane, and points away from the mid-plane on each side (for $\rho > 0$).

Inside ($|x| < d$). Use a pillbox with faces of area A at $\pm x$:

$$2EA = \frac{\rho(2x)A}{\epsilon_0} \Rightarrow E = \frac{\rho x}{\epsilon_0}.$$

Outside ($|x| > d$). The pillbox now encloses the full thickness $2d$:

$$2EA = \frac{\rho(2d)A}{\epsilon_0} \Rightarrow E = \frac{\rho d}{\epsilon_0},$$

which is uniform.

Sketch. $E(x)$ rises linearly from $-\rho d/\epsilon_0$ at $x = -d$, through zero at the centre, to $+\rho d/\epsilon_0$ at $x = +d$, and stays constant beyond. Far away, the slab looks like a sheet with $\sigma = 2\rho d$, whose field $\sigma/2\epsilon_0 = \rho d/\epsilon_0$ agrees. ✓

P14.16

Gaussian sphere of radius $r < R$. By symmetry, $\vec{g}$ is radial and inward, with the same magnitude everywhere on the sphere. The mass enclosed is $M(r/R)^3$:

$$-g \cdot 4\pi r^2 = -4\pi G M \frac{r^3}{R^3} \Rightarrow g = \frac{GM r}{R^3}.$$

At the surface this is $GM/R^2 = g_0$, so $g = g_0 r/R$, exactly the linear result of Chapter 8, here obtained without any integration. Outside, the same method gives $g = GM/r^2$.

P14.17

(a) **Field between the conductors:** $E = 2k\lambda/r$.

$$E(1.0 \text{ mm}) = \frac{2(8.99 \times 10^9)(1.0 \times 10^{-8})}{1.0 \times 10^{-3}} = 1.8 \times 10^5 \text{ N/C}.$$

The largest field is at the surface of the inner conductor, $r = 0.50 \text{ mm}$: $3.6 \times 10^5 \text{ N/C}$.

(b) **Maximum charge.** Set the field at $r = 0.50 \text{ mm}$ equal to the breakdown field:

$$\lambda_{\max} = \frac{E_{\max} r}{2k} = \frac{(2.0 \times 10^7)(5.0 \times 10^{-4})}{2(8.99 \times 10^9)} = 5.6 \times 10^{-7} \text{ C/m} = 0.56 \mu\text{C/m}.$$

(c) **No field outside.** A Gaussian cylinder outside the cable encloses $+\lambda - \lambda = 0$, and by symmetry the field there is radial and uniform over the cylinder. So $\vec{E} = 0$ outside. The signal's field stays inside the cable, and external fields are screened out by the grounded outer conductor. ECG and EEG signals are only microvolts to millivolts, while mains wiring and equipment produce interfering fields. Shielded cables are essential to keep that interference out of the recording.

P14.18

(a) **Total charge of the cloud.** As $r \rightarrow \infty$, e^{-2r/a_0} times any polynomial tends to zero, so the bracket tends to 1 and the enclosed cloud charge tends to $-e$. ✓

(b) At $r = a_0$:

$$Q_{\text{cloud}} = -e [1 - e^{-2}(1 + 2 + 2)] = -e(1 - 5 \times 0.1353) = -0.323 e.$$

Adding the proton, the net enclosed charge is $+e - 0.323e = +0.677e$.

$$E = \frac{k(0.677e)}{a_0^2} = \frac{0.677(8.99 \times 10^9)(1.602 \times 10^{-19})}{(5.29 \times 10^{-11})^2} = 3.5 \times 10^{11} \text{ N/C.}$$

(c) **Far away.** As r grows, the electron cloud encloses almost all of its charge, so the net enclosed charge falls *exponentially* to zero, as e^{-2r/a_0} . By Gauss's law, the field outside a spherically symmetric neutral atom vanishes. Inside the atom, fields of order 10^{11} N/C bind the electron, about 100 000 times the breakdown field of air. This is why neutral atoms barely interact over long distances, and only weakly through induced dipoles at short range.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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